Sales Performance & Discount Profitability Report

Executive Profitability & Discount Analysis (2020–2023)

1. Project Overview

This project extends the existing Sales Dashboard with an Executive Profitability & Discount Analysis covering the period 2020–2023. The analysis evaluates sales performance and product profitability, with a particular focus on the relationship between discount levels and Profit Margin. It aims to identify profitable and loss-making product categories and subcategories, assess profitability across discount levels, and examine changes in key financial indicators over time.

The analytical findings are presented through Power Bi visualisations, with yearly dashboard screenshots documenting the results for each year from 2020 to 2023.

2. Business Objectives

The analysis addresses the following key business questions:

- **Overall Profitability:** What are the annual sales revenue, total profit, and profit margins for each year?
- **Discount Analysis:** How do sales, total profit, and profit margins vary across different discount levels?
- **Product Profitability:** Which product categories and subcategories generate the highest profits, and which are loss-making?
- **Discount Distribution:** How are orders distributed across different discount levels?
- **Year-over-Year Performance:** How do sales performance and profitability change between 2020 and 2023, and how does the relationship between discount levels and profit margins vary over time?

The overall objective was to assess sales profitability, identify loss-making product segments and examine the relationship between discount levels and profit margins. The findings aim to provide data-driven insight to support pricing decisions, discount management, and product portfolio optimisation.

3. Data Preparation

The analysis used two tables from the source dataset: Orders and Products. Data preparation was carried out in Power Query in Microsoft Excel, where the selected fields were reviewed and data types were standardised. The prepared tables were subsequently exported as CSV files and imported into SQL Server, where data exploration, data quality checks and data validation were carried out. The data model and measures were then developed in Microsoft Power BI.

Key data transformation procedures included:

- Reviewing and standardising data types in the fields used for analysis.
- Checking relevant fields for missing or null values and inconsistent entries.
- Identifying duplicate records and repeated identifiers in the source tables.
- Configuring table relationships in Power BI, taking the available key fields, their uniqueness, and relationship cardinality into account.

During the data preparation process, a total of 32 repeated Product IDs were identified, affecting 64 records in the Products table. These duplicate records were removed in Power BI to ensure that each Product ID appeared only once. This enabled the Products table to serve as the one side of a one-to-many (1:N) relationship with the Orders table, where individual Product IDs can be associated with multiple order records.

Following data validation, no missing values were identified in the core analytical fields reviewed, providing a consistent basis for the profitability and discount calculations.

4. Analytical Methodology

This analysis applied descriptive and comparative methods to assess sales performance and profitability during the 2020-2023 reporting period. Sales and profit figures were aggregated by reporting year, discount level, product category, and subcategory.

Performance was assessed using the following Key Performance Indicators (KPIs):

Key Performance Indicator	Measurement Formula & Implementation Logic
Total Sales	SUM(Sales)
Total Profit	SUM(Profit)
Profit Margin	SUM(Profit) / SUM(Sales) Formatted as %
Total Orders	DISTINCTCOUNT(Order ID)

The supporting analytical activities included:

- Comparing total profit and profit margins at different discount levels.
- Examining how orders are distributed among individual discount levels.
- Identifying profitable and loss-making product categories and subcategories.
- Reviewing annual changes in sales, total profit, and profit margins over the 2020–2023 period

5. Dashboard Overview

The Executive Profitability & Discount Analysis dashboard was developed in Microsoft Power BI to provide an overview of sales performance and profitability. It brought together four KPIs, six analytical visualisations, and an interactive year slicer covering the period 2020-2023.

- **1. Executive KPIs:** Display key financial metrics: Total Sales, Total Profit, Profit Margin, and Total Orders.
- **2. Profitability by Category & Discount Level:** A clustered column chart compares Total Profit across product subcategories at different discount levels.
- **3. Discount Tipping Point (Sales vs. Profitability):** A dual-axis combination chart compares Total Sales and Total Profit at different discount levels, illustrating how the two measures vary between discount levels.
- **4. Orders by Discount Level:** A doughnut chart shows how orders are distributed among individual discount levels.
- **5. Profitability & Discount Matrix:** A matrix presents Total Profit by product category, subcategory, and discount level, enabling a more detailed examination of product profitability.
- **6. Performance by Category & Subcategory:** A clustered horizontal bar chart compares Total Sales and Total Profit for individual product categories and subcategories.
- **7. Profit Margin by Discount Level:** A line chart illustrates how Profit Margin varies at individual discount levels.
- **8. Temporal Slicer:** An interactive year slicer allows users to filter dashboard data by individual reporting years from 2020 to 2023.

6. Key Findings

6.1. Overall Financial Performance

The business generated positive Total Profit in each of the four reporting years. Total Sales increased overall from $484,247.49 in 2020 to $733,215.25 in 2023, although they initially declined to $470,532.52 in 2021 before recovering in the following two years.

Total Profit increased consistently throughout the period, rising from $49,543.97 in 2020 to $93,439.26 in 2023. Profit Margin remained positive in all four years, ranging from 10.23% to 13.43%. The highest margin was recorded in 2022, followed by a slight decline to 12.74% in 2023.

6.2. Discount Impact on Profitability

The results showed that Profit Margin generally declined as discount levels increased. Sales made without a discount (0%) achieved the highest margins, while profitability declined as discount levels increased.

The annual Profit Margin values recorded within each discount level over the four-year period ranged as follows:

Discount Level	Annual Profit Margin Range
0%	27.27% to 30.40%
1%–10%	14.73% to 17.14%
11%–20%	11.27% to 13.00%

21%–40%	-19.70% to -10.62%
40%+	-95.72% to -45.60%

Discount levels above 20% were associated with negative Profit Margins in all four reporting years. The lowest Profit Margins were recorded in the highest discount band (40%+), where annual margins ranged from -95.72% to -45.60%.

These findings highlight a strong association between higher discount levels and reduced profitability. However, the descriptive analysis does not establish discounting as the sole cause of the negative financial results.

Note: Order counts by discount level are based on distinct Order IDs. As a single order may contain multiple products sold at different discount levels, the same Order ID can appear in more than one

discount band. Consequently, order counts across discount categories are not mutually exclusive and should not be summed to determine the total number of unique orders.

6.3. Product Profitability & Synthesis with Discount Exposure

The subcategory-level analysis revealed substantial differences in profitability between product groups. While some product lines consistently ranked among the leading profit contributors, others recorded recurring losses or relatively weak financial results.

Top-Performing Subcategories:

- **Accessories:** Ranked among the three highest profit contributors in all four reporting years, securing third place in 2020 and 2022, and second place in 2021 and 2023.
- **Phones:** Ranked first in Total Profit in both 2020 and 2021, and third in 2023, placing among the three leading subcategories in three of the four years.
- **Copiers:** Improved considerably over the reporting period, moving from eleventh place in 2020 to third place in 2021 and first place in both 2022 and 2023.

Least Profitable Subcategories:

- **Tables:** Ranked as the least profitable of all 17 subcategories in every reporting year, consistently recording negative Total Profit. The loss reached $8,140.69 in 2023.
- **Bookcases:** Bookcases remained among the four lowest-ranked subcategories throughout the period, recording negative Total Profit in 2020, 2021, and 2023, with a small positive result in 2022.
- **Supplies:** Recorded negative Total Profit in three consecutive years, from 2021 to 2023.
- **Machines:** Showed a more variable performance, generating positive Total Profit between 2020 and 2022 before falling to the second-lowest position in 2023.

6.4. Category Performance

The category-level comparison revealed substantial differences in financial performance and profitability among the three main product categories. The combined results for the 2020–2023 period are summarised below:

Category	Total Sales	Total Profit	Profit Margin
Technology	$836,154.03	$145,454.95	17.40%
Office Supplies	$719,047.03	$122,490.80	17.04%
Furniture	$741,999.80	$18,451.27	2.49%

Technology generated the highest Total Sales ($836K) and Total Profit ($145.45K), accounting for approximately 50.8% of overall profit. Office Supplies achieved a similarly strong Profit Margin of 17.04%, with Total Profit reaching $122.49K.

In contrast, Furniture generated $742K in Total Sales, comparable to Office Supplies, but contributed only $18.45K in Total Profit. Its Profit Margin of 2.49% was substantially lower than that of the other two categories.

Although Furniture accounted for 32.3% of total sales, it contributed only 6.4% of overall profit. This highlights a considerable difference between its revenue contribution and actual profitability, identifying Furniture as the weakest-performing product category in terms of profit generation.

7. Year-over-Year Analysis

This section compares annual financial performance, order counts, discount-related profitability, and product category results over the 2020–2023 period. It highlights changes in key performance indicators and identifies differences in profitability between product groups over time.

7.1. Overall Business Growth

The business recorded overall growth in sales, Total Profit, and order count between 2020 and 2023, although the year-on-year development varied between individual reporting periods.

Year	Total Sales	Total Profit	Profit Margin	Total Orders
2020	$484,247.49	$49,543.97	10.23%	969
2021	$470,532.52	$61,618.61	13.10%	1,038
2022	$609,205.60	$81,795.18	13.43%	1,315
2023	$733,215.25	$93,439.26	12.74%	1,687

The year-on-year changes in the main performance indicators were as follows:

Performance Metric	2021 vs. 2020	2022 vs. 2021	2023 vs. 2022
Total Sales	-2.83%	+29.47%	+20.36%
Total Profit	+24.37%	+32.74%	+14.24%
Profit Margin	+2.87 pp	+0.33 pp	-0.69 pp
Total Orders	+7.12%	+26.69%	+28.29%

Note: Percentage-point (pp) changes represent differences between the displayed annual Profit Margin values. Percentage changes in Total Sales, Total Profit, and Total Orders represent relative year-on-year changes.

In 2021, Total Sales declined by 2.83%, while Total Profit increased by 24.37%. Consequently, Profit Margin improved from 10.23% to 13.10%, despite the lower sales revenue.

The following year brought stronger results, with Total Sales increasing by 29.47% and Total Profit by 32.74%. Profit Margin reached its highest level of the reporting period at 13.43%.

In 2023, Total Sales continued to grow by 20.36%, while the number of orders increased by 28.29%. However, Total Profit growth slowed to 14.24%, contributing to a 0.69 percentage-point decline in Profit Margin to 12.74%.

Over the entire four-year period, Total Sales increased by 51.41%, Total Profit grew by 88.60%, and Total Orders rose by 74.10%.

7.2. Discount Profitability Trends

The year-on-year comparison of Profit Margin by discount level revealed a consistent relationship between discount depth and profitability throughout the reporting period.

Discount Level	2020	2021	2022	2023
0%	27.27%	30.40%	30.02%	29.86%
1%–10%	14.73%	16.97%	16.29%	17.14%
11%–20%	11.32%	11.31%	11.27%	13.00%
21%–40%	-14.36%	-19.70%	-10.62%	-17.46%
40%+	-45.60%	-89.56%	-95.72%	-92.52%

Note: The 40%+ band represents discount rates exceeding 40%, as an exact 40% discount is included in the preceding 21%–40% category.

Transactions without discounts consistently achieved the highest Profit Margins, ranging from 27.27% to 30.40%. Moderate discount levels (1%–20%) also remained profitable throughout all four years, although their margins were considerably lower.

In contrast, discount levels exceeding 20% were consistently associated with negative Profit Margins. The most severe results occurred in the 40%+ category, where the margin reached -95.72% in 2022 and remained deeply negative at -92.52% in 2023.

These results demonstrate a consistent association between deeper discounts and weaker profitability. However, the observed patterns do not establish a direct causal relationship between discount levels and overall business performance.

7.3. Product Category Performance Trends

The comparison of category-level results between 2020 and 2023 revealed contrasting developments in sales and profitability. While all three categories increased their Total Sales, their contributions to Total Profit followed different trajectories.

Category	Sales 2020	Sales 2023	Sales Change	Profit 2020	Profit 2023	Profit Change
Furniture	$157,192.85	$215,387.27	+37.02%	$5,457.73	$3,018.39	-44.70%
Office Supplies	$151,776.41	$246,097.18	+62.14%	$22,593.42	$39,736.62	+75.88%
Technology	$175,278.23	$271,730.81	+55.03%	$21,492.83	$50,684.26	+135.82%

Margin shift by category highlights structural divergence:

Category	Profit Margin 2020	Profit Margin 2023	Margin Shift
Furniture	3.47%	1.40%	-2.07 pp
Office Supplies	14.89%	16.15%	+1.26 pp
Technology	12.26%	18.65%	+6.39 pp

Technology delivered the strongest improvement in profitability, increasing Total Profit by 135.82% and raising its Profit Margin from 12.26% to 18.65%. It consequently became the leading category in terms of both Total Sales and Total Profit in 2023.

Office Supplies also demonstrated positive development. Total Profit increased by 75.88%, accompanied by an improvement in Profit Margin from 14.89% to 16.15%.

Furniture followed a different trajectory. Although Total Sales increased by 37.02%, Total Profit declined by 44.70%. Its Profit Margin fell by 2.07 percentage points to just 1.40% in 2023.

This contrast highlights the difference between sales growth and profitability. Despite generating substantially higher sales, Furniture achieved weaker financial results, while Technology and Office Supplies converted their sales into profit more effectively.

7.4. Product Subcategory Trends

The annual ranking of product subcategories identified consistent profit contributors as well as those with recurring weak or negative results.

Top three subcategories by Total Profit:

Year	First Place	Second Place	Third Place
2020	Phones ($11,808.01)	Chairs ($6,955.05)	Accessories ($6,402.72)
2021	Phones ($10,398.81)	Accessories ($10,197.28)	Copiers ($9,930.30)
2022	Copiers ($17,742.79)	Binders ($10,215.63)	Accessories ($9,664.29)

2023	Copiers ($25,031.79)	Accessories ($15,672.36)	Phones ($12,849.33)

Three subcategories with the lowest Total Profit:

Year	Lowest Total Profit	Second Lowest	Third Lowest
2020	Tables (-$3,124.04)	Bookcases (-$346.17)	Fasteners ($178.56)
2021	Tables (-$3,509.80)	Bookcases (-$2,755.23)	Supplies (-$24.88)
2022	Tables (-$2,950.94)	Supplies (-$698.96)	Bookcases ($212.47)
2023	Tables (-$8,140.69)	Machines (-$2,869.22)	Supplies (-$955.31)

Note: Subcategories in the second table are ranked by ascending Total Profit. Therefore, the third-lowest position does not necessarily represent a loss, as illustrated by Fasteners in 2020 and Bookcases in 2022.

Accessories consistently ranked among the three leading subcategories in all four years, demonstrating a strong and sustained contribution to overall profitability.

Phones ranked first in both 2020 and 2021 and returned to third place in 2023. Copiers showed particularly strong improvement, moving from third place in 2021 to first place in both 2022 and 2023. Their Total Profit increased from $9,930.30 in 2021 to $25,031.79 in 2023, representing growth of 152.07%.

Tables consistently ranked last among all 17 subcategories, recording negative Total Profit in every reporting year. Their losses reached $8,140.69 in 2023, the largest annual loss recorded within this subcategory during the analysed period.

Bookcases also recorded recurring losses, although they achieved a small positive result in 2022. Supplies generated negative Total Profit in three consecutive years between 2021 and 2023. Machines

displayed a different pattern, remaining profitable between 2020 and 2022 before recording a loss of $2,869.22 in 2023.

7.5. Overall Development Summary

The four-year comparison revealed overall growth in Total Sales, Total Profit, and the number of orders. Between 2020 and 2023, Total Sales increased by 51.41%, while Total Profit grew by 88.60% and Total Orders by 74.10%.

Despite the positive overall development, profitability did not improve consistently throughout the period. Profit Margin peaked at 13.43% in 2022 before declining to 12.74% in 2023, when profit growth also slowed relative to sales growth and the increase in order count.

The analysis highlighted three important findings:

- Discount profitability: Discount levels exceeding 20% were consistently associated with negative Profit Margins throughout all four reporting years.

- Category performance: Technology and Office Supplies generated the strongest profit contributions, while Furniture recorded substantially weaker profitability despite its significant share of Total Sales.

- Subcategory performance: Accessories consistently ranked among the leading profit contributors, whereas Tables remained the least profitable subcategory in every year analysed.

Overall, the findings identify discount management and Furniture category profitability as important areas for further commercial review. They also highlight the importance of monitoring individual subcategories, particularly those with recurring negative profit results.

8. Business Recommendations

The following recommendations are based on the observed profitability patterns across discount levels, product categories, and subcategories.

1. Strengthen governance of discounts exceeding 20%

Introduce stricter approval requirements for discounts above 20%. These discount levels were consistently associated with negative Profit Margins in all four reporting years. Any exceptions should

require a documented business justification and an assessment of their expected impact on profitability.

2. Review profitability in the Furniture category

Review pricing, discount policies, product mix, and cost structures within Furniture. Although Total Sales increased by 37.02% between 2020 and 2023, Total Profit declined by 44.70%, while Profit Margin fell from 3.47% to 1.40%. The category therefore warrants further commercial review.

3. Prioritise the most profitable product categories

Maintain a strong commercial focus on Technology and Office Supplies. Together, these categories generated approximately 93.6% of Total Profit over the reporting period, with Profit Margins of 17.40% and 17.04%, respectively. Their contribution should be considered when setting commercial priorities and evaluating product portfolio performance.

4. Address recurring losses in the Tables subcategory

Review pricing, discount exposure, and product costs within Tables, which recorded negative Total Profit in every reporting year. Its loss reached $8,140.69 in 2023. Further product-level assessment should determine whether pricing or cost adjustments could improve profitability, or whether individual products should be considered for rationalisation.

5. Monitor leading profit-generating subcategories

Continue monitoring Accessories, Copiers, and Phones, which repeatedly ranked among the three leading subcategories by Total Profit. Pay particular attention to maintaining their profitability when reviewing pricing and promotional decisions.

6. Align performance monitoring with profitability

Evaluate commercial performance using Total Sales, Total Profit, Profit Margin, and Total Orders together. Monitor the relationship between discount levels and profitability to ensure that sales growth is not assessed in isolation from its contribution to Total Profit.

9. Limitations and Assumptions

The following limitations should be considered when interpreting the findings:

- Temporal Scope: The analysis covers historical data from 2020 to 2023. External market factors were not assessed.
- Product Identifiers: 32 repeated Product IDs affecting 64 records were removed from the Products table in Power BI to ensure unique identifiers.

- Discount Aggregation: One Order ID may appear in multiple discount levels because an order can contain items with different discounts. Order counts across discount levels are therefore not mutually exclusive.
- Subcategory Scope: The detailed year-on-year comparison focuses on the three highest- and lowest-performing subcategories by Total Profit in each year.
- Financial Scope and Interpretation: The analysis uses the dataset's Sales and Profit fields and excludes indirect costs and overheads. Profit Margin is calculated as Total Profit divided by Total Sales. Observed relationships do not establish causation.